Supplementary Information

Online Resource 1

State ranking, rule consistency and output contracts in medical agent decision components

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S1 Evidence ledger and hierarchy

The completed analysis contains 4,060 responses from 4,068 physical requests and 4,062 normal returns. Supplementary Table S1 reproduces the batch ledger from table1_design_inventory.csv. The 21 saved v7 responses were retained in v8 and counted once in the completed analysis. Two normal v5 returns were excluded from completed analysis. Replotting a response does not create an additional observation.

The primary cross-configuration comparison comprises 1,200 v6/v8/v9 responses (600 Jev, 200 Qwen, 200 GLM, and 200 DeepSeek). Internal batch identifiers are retained here to map the scientific analyses to the frozen collection records. The v4 factorial experiment supplies 600 supporting exploratory responses; v10 and v11 independently contribute 400 and 60 responses. The remaining 1,800 responses belong to earlier exploration in the same study sequence and do not provide independent replications of the later findings.

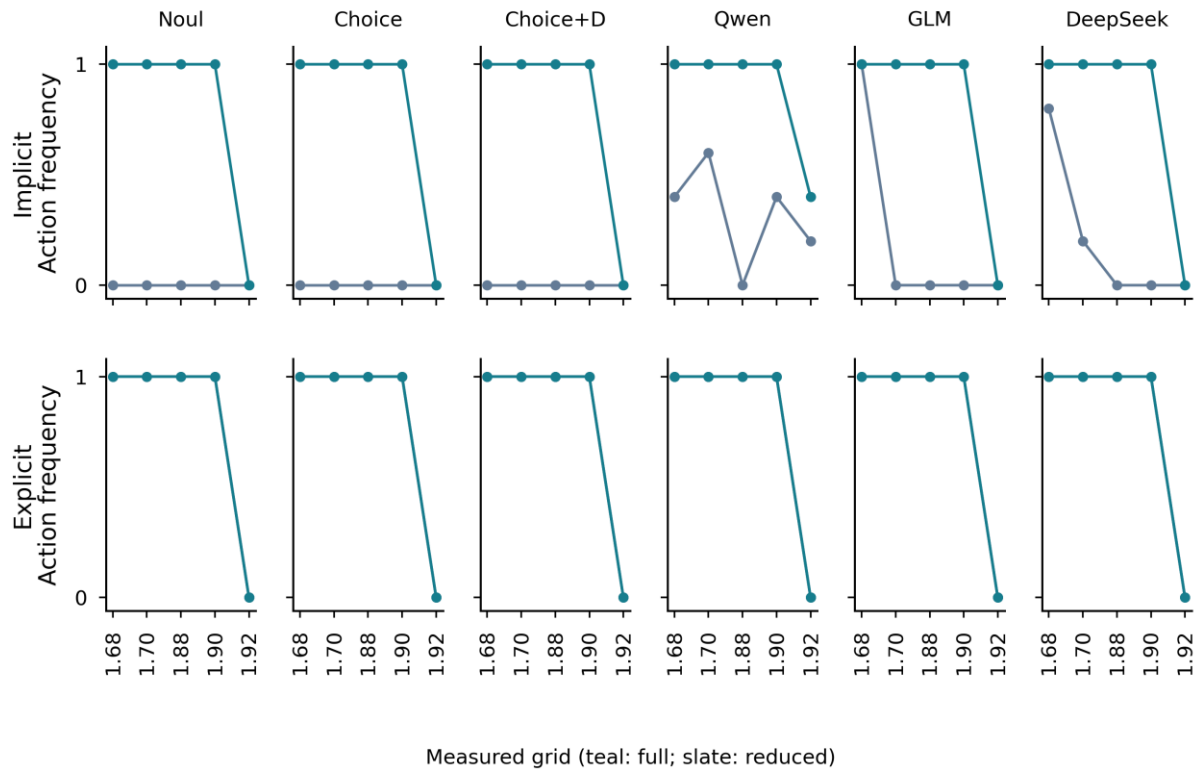
S2 Inputs and checks of the rule references

The Mg and K structures were normalized from MedAgentBench tasks task5_7 and task9_1; the HbA1c structure was derived from task10_1. Each state contains one script-constructed FHIR Observation. These states are not newly retrieved patient records. Main-text Table 2 describes information visibility. Implicit reduced Mg input retains the phrase “if low” without an appended numerical threshold. Full Mg context supplies ranges and dosing that can provide threshold cues. Full K context includes the target of 3.5 mmol/L and replacement/recheck instructions. Both temporal contexts retain the current clock.

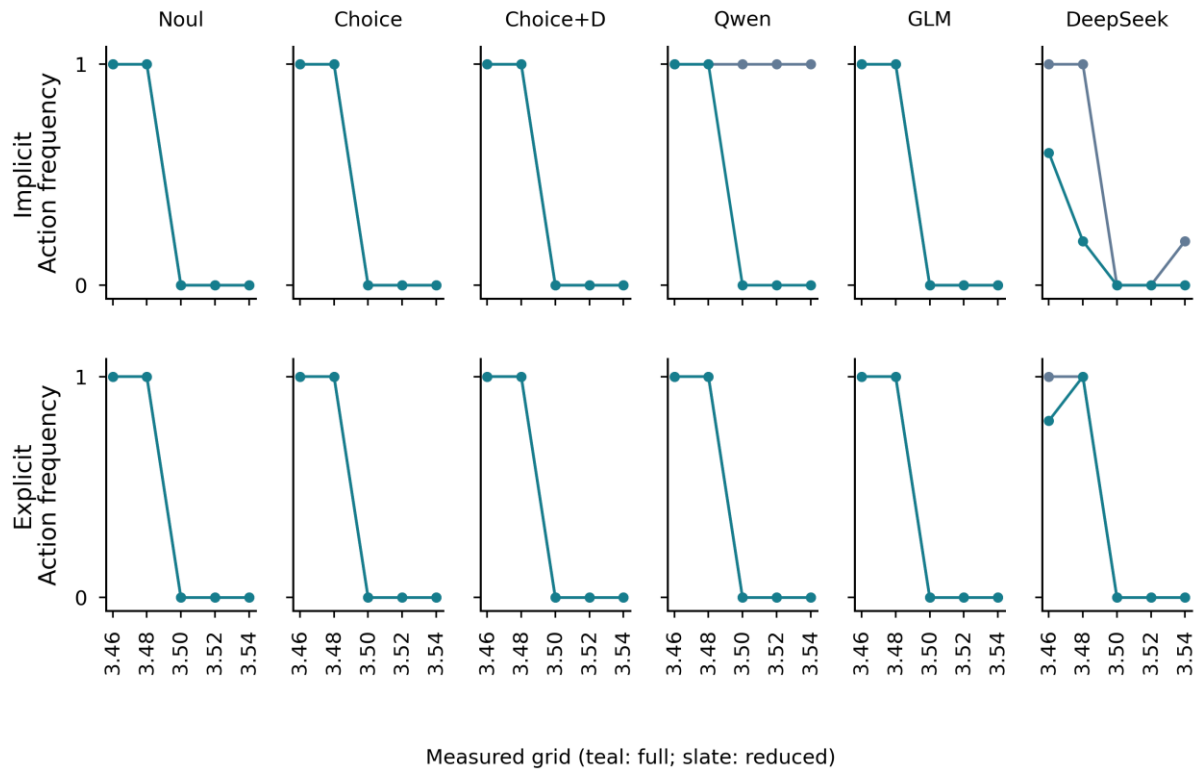
A separate small implementation in analyze_and_plot.py applied $Mg \leq 1.9$, $K < 3.5$, and measurement time strictly before the HbA1c cutoff to the 40 numerical and 20 temporal states. All 60 labels agreed with the saved references; oracle_checks60.csv retains the checks. This procedure detects comparison-operator and equality errors. It is neither an additional model baseline nor a clinical validation.

S3 Complete context strata and state grids

The file all48_context_strata.csv contains 48 task-by-instruction-by-context-by-configuration strata. The file all240_context_cells.csv contains 240 state cells, totaling 1,200 responses. The file context_effects24.csv reports I, C, and M in each condition and the full-minus-reduced differences in percentage points. M retains its valid-output denominator. Action frequencies in the grids are distinct from the CONFIDENCE values generated by chat models.



Supplementary Fig. S1 *Complete Mg state grids. Two instruction modes, two contexts, and six model/probe configurations are shown, with five repeats per state cell. Teal denotes full context and slate reduced context. The vertical axis shows the frequency of the target action. Horizontal positions represent sampled numerical categories; connecting lines do not imply a continuous fitted curve. Jev actions follow the native policy of each probe. Invalid chat outputs are not recoded as target actions*



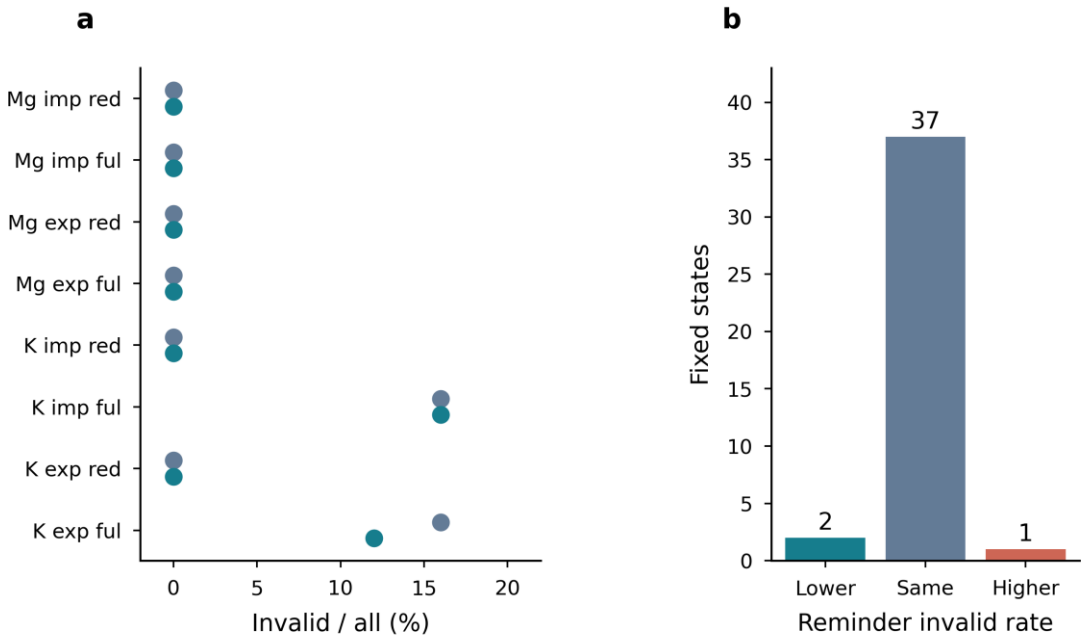
Supplementary Fig. S2 *Complete K state grids. The design matches Supplementary Figure S1, with potassium replacement plus recheck as the target action. Configurations with unchanged or worse outcomes are retained. The underlying state table contains raw action counts, valid-output counts, and strictly correct counts. Action frequency alone does not recover contract validity*

S4 Complete name and range factorial comparison

The file factorial_summary24.csv contains 24 instruction-mode-by-factor-cell-by-probe strata, each with 25 responses. The file all120_factorial_cells.csv contains all 120 state cells. The file factor_effects6.csv reports the range effect at L0 and L1, the name effect at R0 and R1, and their difference in differences. Supplementary Table S2 summarizes the principal contrasts. These are post-hoc descriptive quantities without call-level significance inference. All explicit cells were correct, which does not establish equivalence between the four conditions. L0 changes only severity names, and R0 removes designated numerical-range parentheses; dosing and surrounding text remain.

S5 Label reminder strata and compound label failures

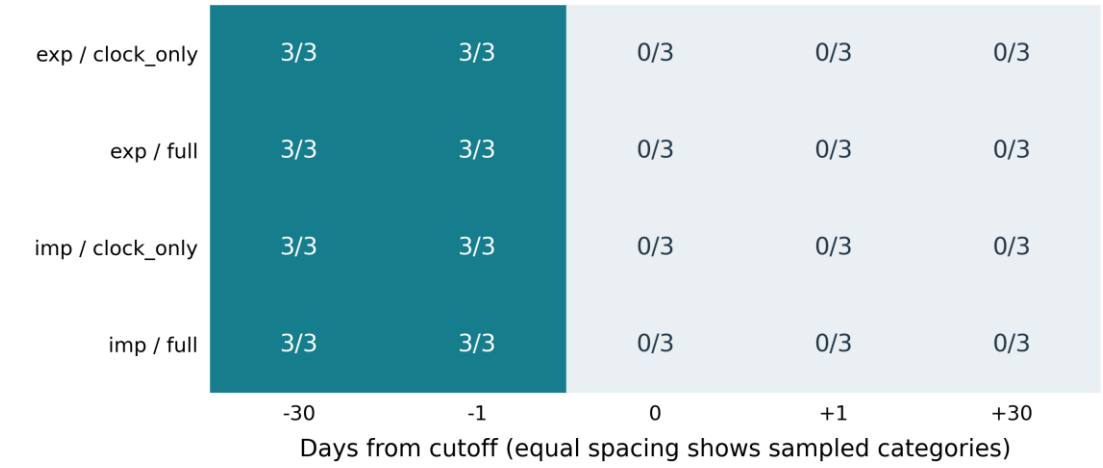
The file v10_endpoint_strata.csv contains the overall results and eight task-by-instruction-by-context strata, with 25 responses per condition in each stratum. All 15 invalid first lines were ACTION: REPLACE_POTASSIUM; the frozen enumeration required REPLACE_POTASSIUM_AND_RECHECK. Labels were not repaired after collection. These invalid outputs are not interpreted as clinical decisions to withhold potassium. The historical field constrained is presented as a textual label reminder and does not denote native schema enforcement.



Supplementary Fig. S3 *Invalid-output locations and changes across all 40 states. The left panel shows invalid-output percentages across eight strata. Slate denotes the original prompt and teal the reminder; each condition contributes 25 responses per stratum. The right panel shows reminder-minus-original invalidity for each state, based on five repeats per condition: two states improve, 37 are unchanged, and one worsens. Supplementary Table S4 retains all three endpoints. The figure is another view of the same responses, not an independent replication*

S6 HbA1c temporal structure

The current clock was fixed at 2023-11-13T10:15:00+00:00 and the cutoff at 2022-11-13T10:15:00+00:00. Measurement strictly before the cutoff triggered a recheck; equality did not. Offsets were −30, −1, 0, +1, and +30 days. Three repeats per state yielded 60 responses. Original/explicit instructions crossed with clock_only/full context each yielded 15/15 strictly correct responses. All 12 equality responses selected no recheck. Missing results, multiple records, and conflicts were outside the tested scope.



Supplementary Fig. S4 *All 20 temporal states. Each cell shows the number of recheck actions out of three repeats. Teal denotes recheck and the light fill no recheck. Columns are discrete sampling categories, not equally spaced time coordinates. All four configurations reach the observed ceiling. The observed transition bracket is [−1, 0] days; no exact boundary is interpolated. The result establishes neither configuration equivalence nor general temporal reasoning reliability*

S7 Sessions and complete paired block sensitivity

The two collection sessions contained 249 and 151 responses on October 16 and October 22, 2026. Original/reminder counts were 125/124 in the first session and 75/76 in the second; invalid counts were 7/6 and 1/1, respectively. The returned alias deepseek-flash was unchanged, while system_fingerprint changed. Both responses in the cross-session block124 were strictly correct. Excluding that block retained 398 responses in 199 complete blocks.



Supplementary Fig. S5 *All responses and complete blocks within sessions. Slate denotes the original prompt and teal the reminder. Response-weighted invalidity is 8/200 versus 7/200 for all responses and 8/199 versus 7/199 after excluding the cross-session block. The latter difference is approximately −0.503 percentage points. State-equal-weighted invalidity is 4.0% versus 3.5% under both specifications, a −0.5-point difference. The two weighting schemes are reported separately. Session coverage was not randomized, so the comparison does not identify a causal backend effect*

S8 Frozen sources and offline reproduction

The research records retain the protocols, prompts, schedules, constructed states, and original model responses. Offline verification replayed all 460 label-reminder and temporal responses with the original strict parsers and checked their schedules, request payloads, state hashes, and returned model identities. Input hashes link the reported analyses to these retained records.

Online Resource 2 (`ESM_2.zip`) provides the aggregate source tables, 600 native Jev score/action records, 400 label-reminder verdict records, 60 temporal verdict records, and portable offline analysis code. Response-level exports retain analysis variables and synthetic state labels; original request bodies, response transcripts, patient-resource text, credentials, and local machine paths are excluded. The archive includes SHA-256 integrity records, a data dictionary, a README, and editable vector figure sources.

Run `python reproduce.py` to verify the included files and recompute the reported context totals, five-state AUROC, factorial contrasts, label-reminder endpoints, temporal counts, and session sensitivity from the supplied analysis variables. The script requires only the Python standard library. The archive also includes `plot_figures.py` and plotting dependencies for rebuilding the published figure designs. Raw-payload and transport verification uses the separately retained research records; it is distinct from reproducing the supplied analysis summaries.

S9 Data access and archive contents

Source tables and analysis code accompany the submission in Online Resource 2. The included variables support the comparisons reported in the article, including unchanged and adverse context effects, all invalid compound labels, and the temporal ceiling. The archive README identifies the source of each table and the granularity of each record. Original benchmark-derived FHIR resources and third-party documents are outside this archive; their reuse terms are governed by their respective sources.

S10 Future study designs

New rule-structure holdouts, native-schema comparisons, multistep evidence acquisition, and a local execution loop remain future work.

Supplementary Table S1. Complete batch ledger

Batch	Physical requests	Normal returns	Analyzed
probe_v1	600	600	600
context_v2	600	600	600
ablation_v3	600	600	600
factorial_v4	600	600	600
llm_v5	4	2	0
qwen_v6	800	800	800
glm_v7	24	21	0
glm_v8	179	179	200
deepseek_v9	201	200	200
contract_ablation_v10	400	400	400
hba1c_temporal_pilot_v11	60	60	60

v8 includes 21 retained v7 responses; they are counted once in analyzed totals. Two normal v5 returns were excluded. Totals are 4,068 physical requests, 4,062 normal returns, and 4,060 analyzed responses.

Supplementary Table S2. Name and range effects in percentage points

Instruction/probe	R@L0	R@L1	L@R0	Difference in differences
Explicit/Noul	0	0	0	0
Explicit/Choice	0	0	0	0
Explicit/Choice+DEFER	0	0	0	0
Implicit/Noul	56	56	0	0
Implicit/Choice	68	64	4	-4
Implicit/Choice+DEFER	48	44	8	-4

R@L0 and R@L1 denote range effects at each name condition; L@R0 denotes the name effect without ranges. The difference in differences is R@L1 minus R@L0.

Supplementary Table S3. All 24 context comparisons

Configuration/task/instruction	C count/25	I count/25	M count/valid
Noul/Mg/imp	5→25	0→0	20/25→0/25
Choice/Mg/imp	5→25	0→0	20/25→0/25
Choice+DEFER/Mg/imp	5→25	0→0	20/25→0/25
Qwen/Mg/imp	11→23	0→0	14/25→2/25
Noul/Mg/exp	25→25	0→0	0/25→0/25
Choice/Mg/exp	25→25	0→0	0/25→0/25
Choice+DEFER/Mg/exp	25→25	0→0	0/25→0/25
Qwen/Mg/exp	25→25	0→0	0/25→0/25
Noul/K/imp	25→25	0→0	0/25→0/25
Choice/K/imp	25→25	0→0	0/25→0/25
Choice+DEFER/K/imp	25→25	0→0	0/25→0/25
Qwen/K/imp	10→25	0→0	15/25→0/25
Noul/K/exp	25→25	0→0	0/25→0/25
Choice/K/exp	25→25	0→0	0/25→0/25
Choice+DEFER/K/exp	25→25	0→0	0/25→0/25
Qwen/K/exp	25→25	0→0	0/25→0/25
GLM-5.3/Mg/imp	10→25	0→0	15/25→0/25
GLM-5.3/Mg/exp	25→25	0→0	0/25→0/25
GLM-5.3/K/imp	25→25	0→0	0/25→0/25
GLM-5.3/K/exp	25→25	0→0	0/25→0/25
DeepSeek/Mg/imp	10→25	0→0	15/25→0/25
DeepSeek/Mg/exp	25→25	0→0	0/25→0/25
DeepSeek/K/imp	24→19	0→6	1/25→0/19
DeepSeek/K/exp	25→24	0→1	0/25→0/24

Arrows indicate reduced → full context. Each condition has 25 responses per stratum. I and C use all responses; M uses valid responses. imp/exp denote implicit/explicit instructions.

Supplementary Table S4. All eight reminder strata

Stratum	Condition	I	C	M
Mg/imp/red	Original	0/25	9/25	16/25
Mg/imp/ful	Original	0/25	25/25	0/25
Mg/exp/red	Original	0/25	25/25	0/25
Mg/exp/ful	Original	0/25	25/25	0/25
K/imp/red	Original	0/25	23/25	2/25
K/imp/ful	Original	4/25	21/25	0/21
K/exp/red	Original	0/25	25/25	0/25
K/exp/ful	Original	4/25	21/25	0/21
Mg/imp/red	Reminder	0/25	10/25	15/25
Mg/imp/ful	Reminder	0/25	25/25	0/25
Mg/exp/red	Reminder	0/25	25/25	0/25
Mg/exp/ful	Reminder	0/25	25/25	0/25
K/imp/red	Reminder	0/25	24/25	1/25
K/imp/ful	Reminder	4/25	21/25	0/21
K/exp/red	Reminder	0/25	25/25	0/25
K/exp/ful	Reminder	3/25	22/25	0/22

I = invalid/all; C = strictly correct/all; M = mismatch/valid. imp/exp denote implicit/explicit instructions and red/ful denote reduced/full context.

Supplementary Table S5. Session sensitivity and weighting

Subset	Condition	I count	Response-weighted %	State-weighted %
all400	Original	8/200	4.0000	4.0
all400	Reminder	7/200	3.5000	3.5
complete398	Original	8/199	4.0201	4.0
complete398	Reminder	7/199	3.5176	3.5

complete398 excludes the two responses in the cross-session block124. Each retained condition contains 199 responses; state-weighted estimates give each of the 40 states equal weight.